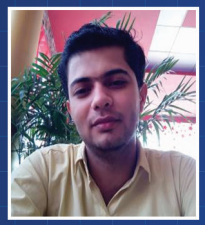
This book is written to provide engineering students and aspiring candidates with relevant information, tricks, tactics, and strategies to optimally achieve their goals. For the students targeting employment in the engineering sector, this book also discusses how to publish a research paper, copyright a computer software, and the entire process of filing a patent in India.

It also mentions optimal steps for candidates planning to go into non-engineering focussed employment such as public sector services, academics, and business administration.

One of the purposes of this book is to make candidates question their thoughts and help them strategise on developing a plan to achieve their end goals.



Ravin Kumar is a curious person blessed to see life as an artist, engineer, researcher, inventor, economist, poet, mathematician, hacker, game developer, and author.

Some of his contributions are mentioned here: a computer language, computer ransomware (and its antivirus), a theorem in Pure Mathematics, addictive auction theory in Theoretical Economics, Mathematical Developments in the Simulation Hypothesis in Physics, cutting edge novel research work in the field of Natural Language Processing, Computer Vision, Deep Learning, a novel algorithm in Quantum Computing, Cyclic scheduling algorithm in Operating Systems, Modified counting sort in Algorithms, and some Patented Inventions, with 50+ software projects and repositories on GitHub, and working experience in product-focused startups in the field of Deep Learning, Machine Learning, Computer Vision, and Natural Language Processing. Along with this, he has done some paintings and related work in the field of Arts, some poetries and is the author of the book "The Engineer's Plan". All these experiences have given him a broader view of life and help understand the world better.

Bachelor of Technology in Computer Science and Engineering, during his graduation (2013-2017), he explored various fields of Computer Sciences to feed his ongoing curiosity and started his career in the field of Artificial Intelligence, Deep Learning in 2017, and since then, he is consistently working in this domain.

His project of experiencing ten different life paths started in August 2013 and got completed on October 31, 2021. The details regarding his project can be found on the link: **https://mr-ravin.github.io**

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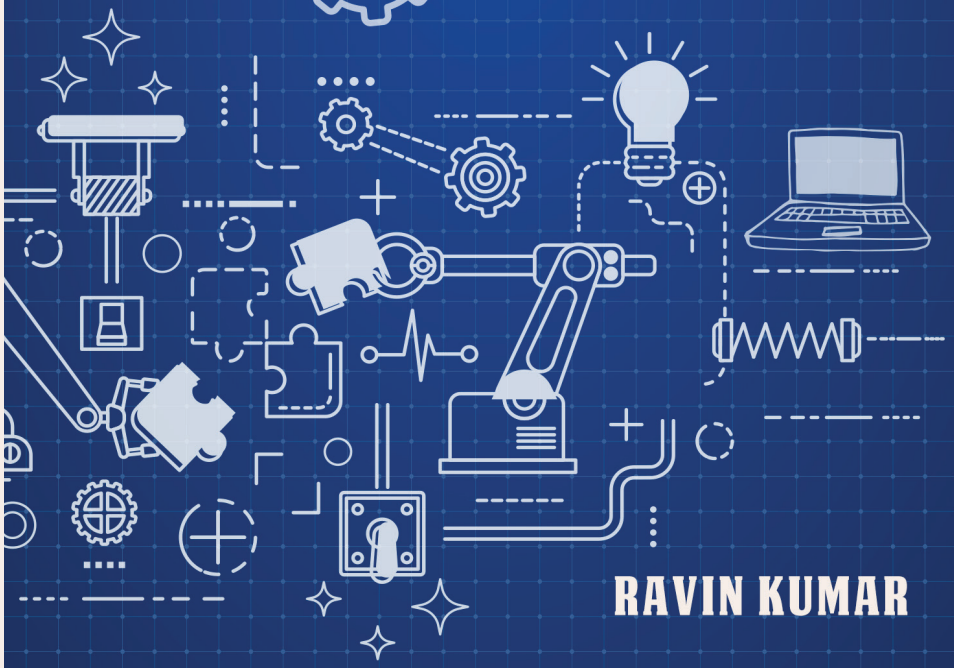

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THE ENGINEER'S PLAN

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Dedication

**This book is dedicated to
my mother Smt. Anuradha,
my little sister Vishu, and my dear wife
Pooja, for their never-ending support
and belief.**

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Acknowledgments

I am grateful for how things have played out, and it looks like even the universe wanted this book to be written. I am also thankful to my friends Harsh and Shubham for their support in making this book a class apart.

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Preface

The objective of this book is to provide relevant information, tricks, and tactics for engineering students. It discusses the optimal steps a student should take to achieve his/her goals.

This book is written in a manner that candidates who are targeting for employment in the engineering or non-engineering sector, both will find the content of this book useful in their journey. It also discusses steps one should take if he/she is targeting post-graduation or will be looking for opportunities in the government sector.

This book is the result of the author's experiment of experiencing ten different life paths; he calls this experiment the "Ten Lives Challenge." The author has tried his best to provide optimal steps for a diverse group of engineering students, each having different dreams, desires, and end-goals but are unfortunately practising the same college curriculum.

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Time And Money

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Human beings have always been curious about time. Some humans had even attempted to conquer it but failed. While some with money are still funding research works focussed on stopping or reversing the ageing process. These humans are trying all these things because they know that time is far more valuable than money. Being a Time Billionaire is far more important than being a Paper Billionaire with less time. This is because, with more time, one can acquire skills to gather money, build status, get fame, or simply enjoy life.

Money, on the other hand, is just a means that humans have mutually agreed to use for transferring the value of their efforts across domains in society.

Another interesting way of looking at money is as a tool to freeze the current time and efforts for future use. This frozen time and effort can then be passed on to the next generation, collected to enhance one's standard of living, or used to attract other humans to contribute their skills and time to achieve one's goal.

At a young age, one should focus on how to efficiently utilise their time. Everyday efforts of acquiring knowledge, building a skill, investing money etc., do compound over time. When one is young, and under normal circumstances, he/she is already a time-billionaire, irrespective of the financial status in the society.

I hope that now it is clear why one should manage their time. Now, let me explain a method which I have found helpful to effectively utilise time. The technique is to create a virtual clock with 24 slots such that its

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1st slot keeps track of 1st to 15th January, the second one keeps track of 16th to 31st January, the 3rd one is for 1st to 14th February, the fourth one is for 15th to February end, and similarly, the 24th one is for 16th to 31st December.

Slot 1	1st Jan - 15th Jan	**Slot 2**	16th Jan - 31st Jan
Slot 3	1st Feb - 15th Feb	**Slot 4**	16th Feb - 28th Feb
Slot 5	1st Mar - 15th Mar	**Slot 6**	16th Mar - 31st Mar
Slot 7	1st Apr - 15th Apr	**Slot 8**	16th Apr - 30th Apr
Slot 9	1st May - 15th May	**Slot 10**	16th May - 31st May
Slot 11	1st Jun - 15th Jun	**Slot 12**	16th Jun - 30th Jun
Slot 13	1st Jul - 15th Jul	**Slot 14**	16th Jul - 31st Jul
Slot 15	1st Aug - 15th Aug	**Slot 16**	16th Aug - 31st Aug
Slot 17	1st Sep - 15th Sep	**Slot 18**	16th Sep - 30th Sep
Slot 19	1st Oct - 15th Oct	**Slot 20**	16th Oct - 31st Oct
Slot 21	1st Nov - 15th Nov	**Slot 22**	16th Nov - 30th Nov
Slot 23	1st Dec - 15th Dec	**Slot 24**	16th Dec - 31st Dec

Now, assign tasks for all these 24 slots to keep track of the progress for the complete human year.

As the performance increases, a person can try to increase the number of cycles in this virtual clock. One can use a virtual clock with two cycles having a total of 48 slots.

Over time, one should aim to achieve utilising a virtual clock having each slot mapped to one hour. In this manner, anyone can slowly but consistently work on increasing their efficiency by mapping their virtual clock to the real world's clock.

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After discussing this system to keep track of our time, let us further understand the relationship between time and money.

In this world, every work has a financial value associated with it. The financial value of a work can be calculated by multiplying the economic value by the amount of time for which the work was performed.

Money (i.e., currency) = Economic value x Time

The economic value is determined by the economic zone providing the money for the efforts performed by the human. The concept of economic zones is a very interesting way of looking at this world from a financial perspective. We will discuss more about economic zones in the coming chapters.

For now, let us discuss how one can freeze time and then pass it on. Assume a situation where the parents worked in an economic zone, and based on the economic value of their skill set and the duration of their work, they received some money. When parents provide money to their children, they are actually adding more time to the child's life. This is because the child has received this money without utilising his/her time or using his/her time to first obtain a skill set to later use to acquire this money. Instead, the child can now use his/her time to enjoy childhood and get better opportunities to acquire a quality skill set with the help of the money received from his/her parents. One such example is the school fee.

Generally, parents who are better in the money aspect of life provide their child better growth opportunities, whereas their counterparts struggle to provide even the basic necessities like food, shelter, and finer healthcare.

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Although one should know that better opportunities do not ensure success in future, they just increase the chances. A student in the top school may not be interested in acquiring the knowledge and will just waste the parent's money.

In the modern era, the internet is the best school where all the necessary quality knowledge is available for almost free. The only requirement is having an interest in the subjects and caring less about fancy degrees, certificates, or ranks. Even if one looks back at the lives of the masters from various fields — Leonardo da Vinci, Isaac Newton, Nikola Tesla, William Shakespeare, Albert Einstein, Amadeus Mozart — they all were focused on understanding and mastering the subjects, rather than trying to get highest marks. Let me put it this way: Do you know who the class topper in which Albert Einstein himself studied was? Well, don't waste your time searching for the answer because the reality is that no one cares about that class topper today. More about education will be discussed in the third chapter.

Hard work is one of the most misused words, people in the middle or lower economic classes often get it confused with the term "work hard". Working hard means working for a longer duration whose economic value is lower, but hard work means working which requires expertise, has demand, and thus, has higher economic value. The advantage of hard work is that one can gather more money even after working for less duration, and the disadvantage of working hard is that even with a longer duration, one will still struggle with the money aspects of life.

Construction workers do physically demanding work that needs extended hours and often are not even

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provided with proper safety at their workplace. But, on the other hand, an architect may design the building, sitting in an air-conditioned room and bill the customer higher than the construction workers. The architect is able to do this because he/she is doing the hard work, i.e., deciding what the final construction will look like and making sure that it is pleasant to the eye. Another example is the difference between a cook and a chef of a restaurant; a cook makes food for the customer's hunger, while a chef makes food for the customer's taste.

I hope now the difference between hard work and working hard is clear. Often what separates hard work from working hard is that hard work requires specialised knowledge or understanding that is high in demand and low in supply. Meanwhile, working hard requires a skill set that is high in supply and low in demand for that economic zone. As a rule of thumb, there will always be more work that requires work-hard than hard-work; because this is how the economy is designed.

Fortunately, one can move from the work-hard mode to the hard-work mode by acquiring a skill set that is high in demand and low in supply. Also, if one stops to improve their skill sets, they might slowly start to move from the hard-work to the work-hard mode. This shifting happens when people in the work-hard mode move to the hard-work mode. It increases the skill set supply, thus decreasing the demand for that skill in their economic zone.

Let me ask one simple question, why does someone go to school? To elaborate further, why do parents send their kids to school, pay the high school fee, and even try their best to send their kids to college, knowing very well that colleges are expensive.

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Parents do work-hard or hard-work to earn money, and then they spend a significant amount of it on their kid's education. They do it because they believe that after getting educated, their kid would acquire knowledge whose economic value would be much higher than the money spent. They, as parents, can then feel proud of raising an amazing kid, which is also one of their duties.

Please note that here the economic values are not just related to financials, but also refers to excellence in the fields which, in turn, improves the quality of life in their economic zone such as being an outstanding scientist, teacher, artist, musician, writer, etc.

Consider a hypothetical situation where a deadly virus outbreak happens in the world, and the scientists and engineers of an economic zone successfully develop a vaccine for this virus. Obviously, that economic zone will have an unfair advantage over the rest. It would attract people with higher financial and skill sets. Hence, as a result, their economy will thrive. All this happened not because of the money the scientists or engineers were making but because of the skill set they possessed. In this case, their skill set is of higher economic value than their earnings. People in that zone would be able to receive vaccines sooner, able to afford them at lower prices, leading to sooner re-opening of the schools, factories, and the whole economy of that zone.

So, one goes to school and gives his/her time and parents money to the school in order to get knowledge whose economic value is high. But in the case of one going to a cinema hall to watch a movie, one needs to pay money, give his/her time to watch it, and they will probably also buy popcorn and cold drinks. Isn't it a terrible transaction for the customer? There are also

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good quality movies that can genuinely add value to one's life, but the quantity of such movies released every year is minimal.

Now let's take an example of sports. Instead of playing their favourite sports on the field, people often just watch the matches on their tv or smartphones, having pizza, burgers, French fries, and cold drinks. This is a sad reality of today's world. Instead of doing the physical activity themselves, people just sit on the couch, watching others play the game while they enjoy their fast food and become fat!!! The problem here is not about watching the game but about not playing it. The players get all the physical and financial benefits and social recognition, while the viewers get nothing.

One should always keep their physical fitness in check as there is no way known to us to extend our lifespan by just using money. Healthy food and a good lifestyle are the keys to living a quality life. Also, don't get tricked by advertisements into buying food, drinks, and snacks that harm the human body, and some even have non-recoverable side effects.

For a more balanced life, one should take care of the physical as well as the mental fitness. Think about it, what's the point of doing hard-work or working-hard if one will give all of the life earnings to the hospitals?

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Economic Zones

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Our world can be seen as a collection of multiple economic zones. Sometimes, even a whole country can be present inside a single economic zone, while in other cases, it might also happen that a country contains multiple economic zones within its boundaries.

Economic zones are areas with different market needs or having different demand and supply requirements. Those demands may be due to cultural, regional, geographical, or even economical viability. For example, geographical regions where the fishes are consumed more are largely located near the water bodies — oceans, seas, etc. — and slowly, over time, the economics and human habits gave rise to a culture and established customs of the locals.

When it comes to determining the amount of money one would get for their work depends not just on the work duration but also on the economic zone from where that money will be collected.

Understanding economic zones is very important, especially when it comes to deciding the field of graduation. To better understand this, let's take an example: One person by birth is highly skilled in Baseball but is born in India; similarly, let's assume another person who is highly skilled in Cricket but is born in the USA. Both will suffer on an economic scale as Baseball is not a popular sport in India, and Cricket is not popular in the USA. So, if they want to increase their economic value, either they would have to switch to a sport with higher popularity in their current

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